

1 Machining Center / Order Pool

Purpose

You can use the functionality "Machining center/Order pool" on the AIP to process the operations, which are logged on in parallel, in a specified sequence.

Requirements

If you use AIP 8.2, you require at least version AIP 8.2.1.26 or service pack 13 (2018).

If the machine data collection is run via a stand-alone PCC ("central MDE"), you require at least service pack 13 (2018) and AIP 8.2.

Check in the [Dynamic dialog configuration](#), that the dialogs NC_AN and NC_AB are existing and activated. If the dynamic dialog NC_AN and NC_AB are not available, proceed as described in chapter [Load dynamic dialogs](#).

If required, execute the configurations described in chapter [AIP configuration](#).

Activation

You enable the functionality "Machining center" on the MOC in the machine/workplace configuration:

Go to: *Master data* → *Workplace/resource configuration* → Tab *Workplace configuration* → *Workplace category=J* <*Machining center*>



The functionality "Machining center" includes the specific posting of durations. For this reason, you must set the option *Posting of machine time with operations logged on simultaneously* to "N" (also included in the tab workplace configuration).

This functionality is also available on the AIP 8.2, if the PCC is run stand-alone ("central MDE").

Posting procedure and posting

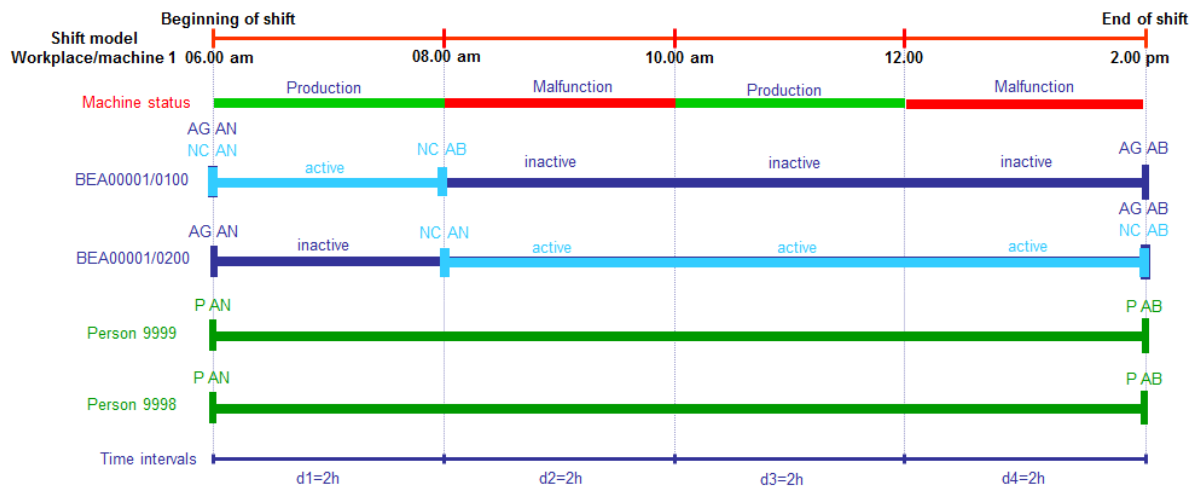
The operator logs on the different operations to the respective machine. Further persons can optionally log on.

One operation is processed at a point in time X. If the machine is in status "Production", the duration and the complete labor utilization is posted to the active OP. If no operation is active, the system does not record any duration for an order. In case of a machine failure, the duration and the labor utilization are proportionally posted to all operations.

If the system records automatic quantities for this machine, the quantities and the production times are posted the same way: the automatic quantities are only posted to the active operation and for the persons logged on to this operation. Manual quantities are assigned to the operation to which the quantities are posted.

Machining Center with Order Pool

Four operations are logged on to the machine at the same time!



Postings in HYDRA

Order
OP BEA00001/0100 (U record)
06.00 - 14.00
⇒Duration: 2,0 hrs.
⇒Labor duration: 8,0 hours

Order
OP BEA00001/0300 (U record)
06.00 - 14.00
⇒Duration: 6,0 hrs.
⇒Labor duration: 8,0 hours

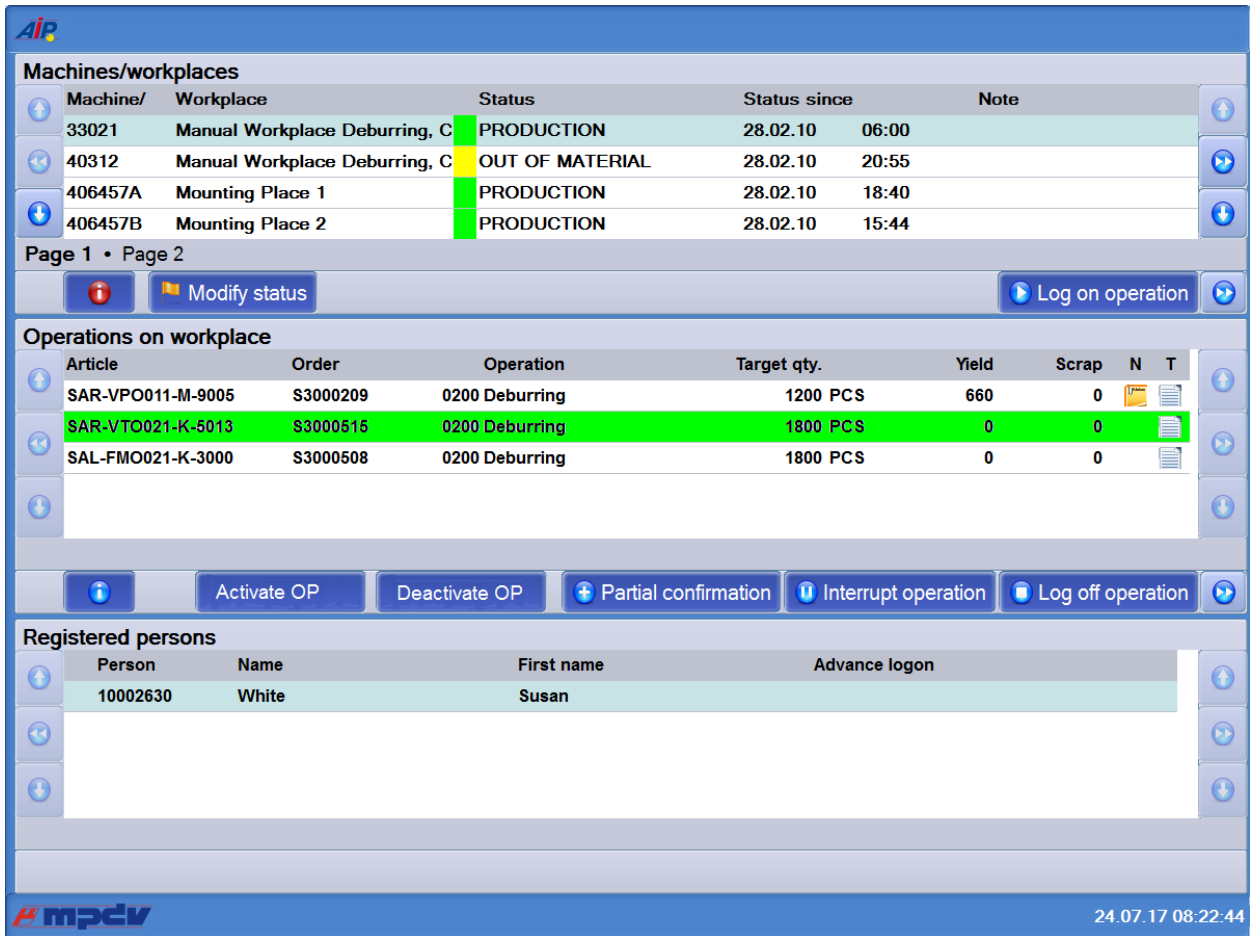
Also a HYDRA group workplace can be a machining center.

You cannot create merged operations for a machining center, because the production times and downtimes are always posted proportionally. This processing of postings contradicts the processing of postings for a machining center.



The posting events "Activate operation" and "Deactivate operation" are displayed in the event maintenance (events NC_AN, NC_AB).

Functions on the terminal AIP 8.1



The screenshot displays the AIP 8.1 terminal interface. The top section, 'Machines/workplaces', contains a table with columns: Machine/, Workplace, Status, Status since, and Note. The second section, 'Operations on workplace', contains a table with columns: Article, Order, Operation, Target qty., Yield, Scrap, N, and T. The third section, 'Registered persons', contains a table with columns: Person, Name, First name, and Advance logon. The interface includes navigation buttons (up, down, left, right) and action buttons (Modify status, Log on operation, Activate OP, Deactivate OP, Partial confirmation, Interrupt operation, Log off operation). The bottom status bar shows the date and time: 24.07.17 08:22:44.

Machine/	Workplace	Status	Status since	Note
33021	Manual Workplace Deburring, C	PRODUCTION	28.02.10 06:00	
40312	Manual Workplace Deburring, C	OUT OF MATERIAL	28.02.10 20:55	
406457A	Mounting Place 1	PRODUCTION	28.02.10 18:40	
406457B	Mounting Place 2	PRODUCTION	28.02.10 15:44	

Page 1 • Page 2

Article	Order	Operation	Target qty.	Yield	Scrap	N	T
SAR-VPO011-M-9005	S3000209	0200 Deburring	1200 PCS	660	0		
SAR-VTO021-K-5013	S3000515	0200 Deburring	1800 PCS	0	0		
SAL-FMO021-K-3000	S3000508	0200 Deburring	1800 PCS	0	0		

Person	Name	First name	Advance logon
10002630	White	Susan	

24.07.17 08:22:44

The list "Operations on workplace" of the main view shows all operations logged on. If the order list is configured accordingly, the operation currently active is colored (green).

In the dialog "Activate OP", the operation selected in the main view is used and preassigned.

In the dialog "Deactivate OP", the active operation is automatically set.

Functions on the terminal AIP 8.2

Main view

The list "Operations on workplace" of the main view shows all operations logged on. A green frame highlights the active operation. The MES order number of the inactive operations is displayed in gray.

AIP																																	
MDEBEARB production 326 0	Workplace/machine																																
MDE99 no order 936 0	<table border="1"> <tr> <td>Workplace/machine</td> <td>Short name</td> <td>Group</td> <td>Status</td> </tr> <tr> <td>MDEBEARB</td> <td></td> <td>DS100</td> <td>production</td> </tr> <tr> <td>Cycles</td> <td>312</td> <td>Start</td> <td>Duration [hrs:min]</td> </tr> <tr> <td></td> <td></td> <td>7/20/2018 4:17 PM</td> <td>0:00</td> </tr> <tr> <td>Target cycle</td> <td>22,000</td> <td>Actual cycle</td> <td>Yield</td> </tr> <tr> <td></td> <td></td> <td>0,000</td> <td>326</td> </tr> <tr> <td></td> <td></td> <td></td> <td>Scrap</td> </tr> <tr> <td></td> <td></td> <td></td> <td>0</td> </tr> </table>	Workplace/machine	Short name	Group	Status	MDEBEARB		DS100	production	Cycles	312	Start	Duration [hrs:min]			7/20/2018 4:17 PM	0:00	Target cycle	22,000	Actual cycle	Yield			0,000	326				Scrap				0
Workplace/machine	Short name	Group	Status																														
MDEBEARB		DS100	production																														
Cycles	312	Start	Duration [hrs:min]																														
		7/20/2018 4:17 PM	0:00																														
Target cycle	22,000	Actual cycle	Yield																														
		0,000	326																														
			Scrap																														
			0																														
Operations																																	
+	<table border="1"> <tr> <td>MES order number</td> <td>MES order number</td> </tr> <tr> <td>DS1000090011</td> <td>ML0000010030</td> </tr> <tr> <td>R-1- Имя операции Test LOG</td> <td>S-3- Имя операции</td> </tr> <tr> <td>Article</td> <td>Article</td> </tr> <tr> <td>109747-00</td> <td>Test Artikel ML ATK 3</td> </tr> <tr> <td>Quantities (target/yield/scrap)</td> <td>Quantities (target/yield/scrap)</td> </tr> <tr> <td>40000000 / 4773 / 1383</td> <td>20000 / 13824 / 10812</td> </tr> </table>	MES order number	MES order number	DS1000090011	ML0000010030	R-1- Имя операции Test LOG	S-3- Имя операции	Article	Article	109747-00	Test Artikel ML ATK 3	Quantities (target/yield/scrap)	Quantities (target/yield/scrap)	40000000 / 4773 / 1383	20000 / 13824 / 10812																		
MES order number	MES order number																																
DS1000090011	ML0000010030																																
R-1- Имя операции Test LOG	S-3- Имя операции																																
Article	Article																																
109747-00	Test Artikel ML ATK 3																																
Quantities (target/yield/scrap)	Quantities (target/yield/scrap)																																
40000000 / 4773 / 1383	20000 / 13824 / 10812																																
Staff logged on																																	
+																																	

8.2.1.26 Server:win2008-7.0 TNR:149 20.07.18 16:17:59

Detail view of the operation

To activate or deactivate an operation, just click on the operation. In the detail view of the operation, the section "Operation" shows if the operation is active (green) or inactive (gray).

AIP																																	
< back	Workplace/machine																																
Confirm partially	<table border="1"> <tr> <td>Workplace/machine</td> <td>Short name</td> <td>Group</td> <td>Status</td> </tr> <tr> <td>MDEBEARB</td> <td></td> <td>DS100</td> <td>production</td> </tr> <tr> <td>Cycles</td> <td>312</td> <td>Start</td> <td>Duration [hrs:min]</td> </tr> <tr> <td></td> <td></td> <td>7/20/2018 4:12 PM</td> <td>0:01</td> </tr> <tr> <td>Target cycle</td> <td>22,000</td> <td>Actual cycle</td> <td>Yield</td> </tr> <tr> <td></td> <td></td> <td>0,000</td> <td>326</td> </tr> <tr> <td></td> <td></td> <td></td> <td>Scrap</td> </tr> <tr> <td></td> <td></td> <td></td> <td>0</td> </tr> </table>	Workplace/machine	Short name	Group	Status	MDEBEARB		DS100	production	Cycles	312	Start	Duration [hrs:min]			7/20/2018 4:12 PM	0:01	Target cycle	22,000	Actual cycle	Yield			0,000	326				Scrap				0
Workplace/machine	Short name	Group	Status																														
MDEBEARB		DS100	production																														
Cycles	312	Start	Duration [hrs:min]																														
		7/20/2018 4:12 PM	0:01																														
Target cycle	22,000	Actual cycle	Yield																														
		0,000	326																														
			Scrap																														
			0																														
Interrupt	Operation																																
Log off	<table border="1"> <tr> <td>MES order number</td> <td>Quantities (target/yield/scrap)</td> <td>Comments</td> <td>State</td> </tr> <tr> <td>DS1000090011</td> <td>40000000 / 4773 / 1383</td> <td></td> <td>Active</td> </tr> <tr> <td>R-1- Имя операции Test LOG</td> <td>0%</td> <td></td> <td></td> </tr> <tr> <td>Article</td> <td>Since logon (target/yield/deviation)</td> <td>Planned duration</td> <td>Partitioning</td> </tr> <tr> <td>109747-00</td> <td>32 / 144 / 354%</td> <td>0:00</td> <td>1</td> </tr> <tr> <td>Test 22222220010 Bezeichnung</td> <td></td> <td></td> <td></td> </tr> </table>	MES order number	Quantities (target/yield/scrap)	Comments	State	DS1000090011	40000000 / 4773 / 1383		Active	R-1- Имя операции Test LOG	0%			Article	Since logon (target/yield/deviation)	Planned duration	Partitioning	109747-00	32 / 144 / 354%	0:00	1	Test 22222220010 Bezeichnung											
MES order number	Quantities (target/yield/scrap)	Comments	State																														
DS1000090011	40000000 / 4773 / 1383		Active																														
R-1- Имя операции Test LOG	0%																																
Article	Since logon (target/yield/deviation)	Planned duration	Partitioning																														
109747-00	32 / 144 / 354%	0:00	1																														
Test 22222220010 Bezeichnung																																	
Change target quantity	Staff logged on																																
Change partitioning																																	
BDE comments																																	
Activate OP																																	
Deactivate OP																																	

8.2.1.26 Server:win2008-7.0 TNR:149 20.07.18 16:13:31

You can use the buttons on the left hand side to "Activate operation" or "Deactivate operation".

Dialog "Activate operation"

In the dialog "Activate operation", the selected workplace is pre-assigned. You can select the operation in the list that you want to activate. If an operation is currently active, this operation is shown in green until you select the row of this operation.

Activate operation

Workplace MDEBEARB

Filter

Article	Order	Operation	Target qty.	Yield	Scrap	N
109747-00	DS100009	0011 R-1- Имя операции Test L...	40000000 ST	4773	1383	
Test Artikel ML ATK 3	ML000001	0030 S-3- Имя операции	20000 ST	13824	10812	

Operation No.

8.2.1.26 Server:win2008-7.0 TNR:149 20.07.18 16:24:27

Dialog "Deactivate operation"

In the dialog "Deactivate operation", the selected workplace and the selected operation are pre-assigned.

Deactivate operation

Workplace MDEBEARB
Operation DS1000090011

8.2.1.26 Server:win2008-7.0 TNR:149 20.07.18 16:22:58

Machine connection

To connect a machine, you must coordinate all details with the customer or the machine manufacturer. A subsequent customization is required. The following details must be discussed:

Details of technical communication.

- Examples: OPC-UA, File-Interface, UMPS, etc.
- Is the PCC run in stand-alone operation? (Note: only possible if you use AIP 8.2)

Details of logical communication.

- Can the machine control provide the data listed in the table below? If no:
 - How can the system identify this information?
 - Do you have to make extensions in the data collection dialogs?
- Which format does the data provided have?
- Does the machine control send a command to activate an operation **and** then to deactivate an operation or does the machine control only send a command to activate an operation?

Commands to activate or deactivate an operation

Field ID	Description
DLG	Dialog/record ID: always: „NC_AN“ – activate OP. „NC_AB“ – deactivate OP.
MNR	Workplace/machine number according to configuration
ANR	MES order number (order/OP)

Beforehand, you must manually log on the operations on the terminal.

Identification of the MES order number

If the machine control cannot pass the MES order number, a specific customization is required to add the MES order number (ANR) to the command to activate or deactivate an operation.

Load dynamic dialogs

Execute the following activities on the HYDRA server in the system directory.

1. Save the existing dialog configuration:

UNIX systems (run at the server prompt in the HYDRA directory):

```
hydlgcfg.out DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg
```

Windows systems (Execute in a DOS window in the HYDRA directory):

```
hydlgcfg.exe DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg
```

2. Now load the new dialog configurations using the following command:

UNIX systems:

```
hymw.out -u9999 -b  
db_sql/aip_bde_bea.dlg
```

Windows systems:

```
hymw.exe -u9999 -b  
db_sql\aip_bde_bea.dlg
```

Please note:

If UNIX systems are used, you must enter the file name in lower case letters, as HYDRA Install converts all file names into lower case letters during installation.

3. Activate the new dynamic dialogs:

UNIX systems:

```
hydialog.scr AIPTNR 0
```

Windows systems:

```
sh.exe hydialog.scr AIPTNR 0
```

Note:

This command activates the default dialogs. If terminal or terminal group-specific dialogs are used on the system, they must be customized by an MPDV consultant.

4. Execute configurations described in chapter "Activation" or the "Configuration AIP". If terminal or terminal group-specific dialogs are used on the system, the settings should ideally be made with the support of an MPDV consultant.
5. Start the terminal and test the function.

AIP configuration

Display of the active operation on the terminal AIP 8.1 (if required)

Older installations of the AIP 8.1 can still include the following configuration. Make the following settings in the ctaipplay.ini to highlight the currently active operation on the terminal:

```
[order list]  
...  
GRID_COLOR=clSilver  
GRID_BACKGROUND=clWhite  
EXAMINE_SCANEXPR1=AKTIV=J  
EXAMINE_SCANEXPR2=AKTIV=N  
EXAMINE_SCANCOLOR1=clBlack  
EXAMINE_SCANBKEXPR1=AKTIV=J  
EXAMINE_SCANBKEXPR2=AKTIV=N  
EXAMINE_SCANBKCOLOR1=clLime  
EXAMINE_SCANBKCOLOR2=clWhite  
...
```

Configuration of the buttons on the terminal AIP 8.1 (if required)

Older installations of the AIP 8.1 can still include the following configuration. Make the following settings in the ctaipbut.ini to show the buttons Activate and Deactivate on the terminal. You can position the buttons anywhere as long as the existing buttons are respected. You must ensure that the buttons are entered in the required section of the file ctaipbut.ini and that numbers are consecutive.

Example:

```
[ANR-ALL-Page4]
1=NC_AN,L,Activate OP
2=NC_AB,L,Deactivate OP
```

Automatic deactivation of active operation

Parameter in ctaip.ini:

```
[System]
NC_AUTO_DEACTIVATE=Y
```

Use this parameter to disable the plausibility function, which ensures that only one operation can be active. At the same time, the following automatic is active when you activate an operation:

The currently (previously) active operation is first deactivated. Then the new operation is activated. Note: The system does not perform these processing steps in one transaction. The system also deactivates the previously active operation if the new operation cannot be activated because of a plausibility error.